B. AND Reconstruction

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1100 | Tags: bitmasks, constructive algorithms, greedy | https://codeforces.com/problemset/problem/1991/B

You are given an array b of n - 1 integers.

An array a of n integers is called *good* if $b_i = a_i \& a_{i+1}$ for $1 \le i \le n$-1, where & denotes the bitwise AND operator.

Construct a good array, or report that no good arrays exist.

Input

Each test contains multiple test cases. The first line contains a single integer t ($1 \le t \le 10^4$) — the number of test cases. The description of test cases follows.

The first line of each test case contains a single integer n ($2 \le n \le 10^5$) — the length of the array a.

The second line of each test case contains n - 1 integers $b_1, b_2, \ldots, b_{n-1}$ ($0 \le b_i$ lt 2^{30}) — the elements of the array b.

It is guaranteed that the sum of n over all test cases does not exceed 10^5.

Output

For each test case, output a single integer -1 if no good arrays exist.

Otherwise, output n space-separated integers $a_1, a_2, \ldots, a_n$ ($0 \le a_i$ lt 2^{30}) — the elements of a good array a.

If there are multiple solutions, you may output any of them.

Example

Input
4 2 1 3 2 0 4 1 2 3 5 3 5 4 2

Output
5 3 3 2 1 -1 3 7 5 6 3

Note

In the first test case, $b = [1]$. A possible good array is $a=[5, 3]$, because $a_1 \,\&\, a_2 = 5 \,\&\, 3 = 1 = b_1$.

In the second test case, $b = [2, 0]$. A possible good array is $a=[3, 2, 1]$, because $a_1 \,\&\, a_2 = 3 \,\&\, 2 = 2 = b_1$ and $a_2 \,\&\, a_3 = 2 \,\&\, 1 = 0 = b_2$.

In the third test case, $b = [1, 2, 3]$. It can be shown that no good arrays exist, so the output is -1.

In the fourth test case, $b = [3, 5, 4, 2]$. A possible good array is $a=[3, 7, 5, 6, 3]$.